

NIFTY 100 INSTITUTIONAL ANALYST GUIDE & SYSTEM MANUAL

A Comprehensive Technical, Mathematical, and Operational Reference for Quantitative Equity Research, Fundamental Intelligence, and REST API Integration

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1. System Architecture & Relational Schema

The Bluestock Nifty 100 Financial Intelligence Engine is built on a modular three-tier architecture comprising an ETL Data Ingestion Engine, a Relational SQLite Database with strict B-Tree indexing, a high-throughput FastAPI REST Server, and an analytical Streamlit dashboard.

Relational Database Schema (10 Core Tables)

Table Name	Primary Key	Row Count	Description
companies	company_id	92	Master metadata (ticker, sector, industry, shares)
profitandloss	(company_id, year)	857	Income statement line items (Sales, OPM, PAT, EPS)
balancesheet	(company_id, year)	857	Balance sheet items (Equity, Reserves, Borrowings, FA)
cashflow	(company_id, year)	857	Cash flow statement (CFO, CFI, CFF, Net Cash)
financial_ratios	(company_id, year)	857	Computed KPIs (ROE, ROCE, D/E, ICR, Asset TO)
cashflow_intelligence	company_id	92	CFO Quality Score, CapEx Intensity, Allocation Pattern
nlp_pros_cons	(company_id, item_type, id)	460+	Extracted qualitative strengths, risks, confidence
ml_cluster_labels	company_id	92	KMeans archetype cluster, PCA coordinates, distance
shareholding_pattern	(company_id, quarter)	368+	Promoter, FII, DII, Public equity ownership stakes
corporate_actions	id	240+	Dividends, bonus issues, stock splits, AGM notices

Data Flow & Integrity Guarantees

- **Single Source of Truth:** SQLite file nifty100.db initialized with WAL (Write-Ahead Logging) and foreign key constraints.
- **Read-Only ETL Pipeline:** Raw Excel workbooks in data/raw/ are read-only and never overwritten during transformations.
- **Determinism:** Random seed 42 enforced across all stochastic ML and clustering modules for reproducible execution.

2. Interactive Dashboard Navigation Guide

The Streamlit multi-page frontend (dashboard/app.py) provides institutional analysts with responsive visualizations, interactive data filters, and dynamic report generation capabilities across 7 specialized pages.

Page Name	File Path	Key Functionalities & Visuals
Home Overview	pages/01_home.py	Market map, sector distribution, top gainers, portfolio aggregate KPIs
Company Deep-Dive	pages/02_company.py	10-year financials, DuPont analysis, interactive interactive Plotly charts
Cash Flow Intel	pages/03_cashflow.py	CFO Quality scores, CapEx intensity scatter, capital allocation matrices
Equity Screener	pages/04_screener.py	Multi-metric slider filters (ROE, ROCE, D/E, Margin) with CSV export
Peer Comparison	pages/05_peer.py	Cross-company benchmarking, percentile radars, sector median delta
Valuation Models	pages/06_valuation.py	DCF calculator, Graham Number, Peter Lynch fair value, Reverse DCF
Reports Hub	pages/07_reports.py	One-click PDF downloads for 92 tearsheets and 11 sector profiles

Key User Interactions & Workflow

- Ticker Search:** Type any of the 92 NSE ticker symbols in the sidebar selector to instantly load complete historical profiles.
- Dynamic Sliders:** Filter stocks in real time using 8 fundamental financial criteria with instantaneous table updates.
- Instant PDF Generation:** Download institutional 2-page Tearsheets with pre-rendered financial charts and KPI tables.

3. Quantitative Equity Screener & Valuation Engines

The valuation and screening suite in `src/analytics/valuation.py` and `src/analytics/screener.py` equips analysts with robust quantitative models to estimate intrinsic fair values and margin of safety.

Valuation Models Implemented

Model Name	Underlying Formula	Analyst Use Case
Discounted Cash Flow (DCF)	$\text{Sum}(\text{FCFF} / (1+\text{WACC})^t) + \text{TV}/(1+\text{WACC})^n$	Cash-generative industrials and IT services with stable CFO
Graham Fair Value	$\text{Sqrt}(22.5 * \text{EPS} * \text{BVPS})$	Conservative value screen for asset-heavy dividend payers
Peter Lynch Fair Value	$\text{PEG} = (\text{P/E}) / \text{CAGR}_{5Y} = 1.0$ Fair Value	Growth-at-a-Reasonable-Price (GARP) screening
Reverse DCF Model	Implied FCF Growth = Solve for Price(WACC, g)	Assessing whether current stock price reflects realistic growth
Historical PE Multiple Band	Price = LTM EPS * Historical (Mean / +/- 1 Std) PE	Mean-reversion cyclical valuation benchmarking

Margin of Safety Interpretation

- **Undervalued (> 20% Discount):** Current market price offers substantial cushion against analytical estimation error.
- **Fairly Valued (-10% to +10%):** Trading close to computed intrinsic benchmark; driven by earnings momentum.
- **Overvalued (> 25% Premium):** High growth expectations already priced in; vulnerability to earnings downgrades.

4. Cash Flow Quality & Capital Allocation Intelligence

Reported net profits can be distorted by accruals and aggressive revenue recognition. The Cash Flow Intelligence module (src/analytics/cashflow_kpis.py) conducts forensics on earnings quality and capital stewardship.

CFO Quality Classification Framework

Classification	Criteria (CFO / PAT Ratio)	Quality Grade	Economic Meaning
High Quality	CFO / PAT ≥ 1.0 in 8+ of 10 years	Grade A (Top)	Realized cash earnings exceed reported book net profit
Moderate Quality	CFO / PAT ≥ 0.8 in 5-7 of 10 years	Grade B	Working capital cycles absorb temporary operating cash
Low Quality	CFO / PAT < 0.8 in > 5 years	Grade C (Flag)	High accruals; risk of receivables bloating or inventory pileup
Negative CFO	Cumulative 3-year CFO < 0	Grade D (Severe)	Cash-burning operational profile; requires external debt/equity

Capital Allocation Archetypes

- Aggressive Re-investor:** CapEx / CFO $> 70\%$, Dividend Payout $< 15\%$. Directs majority of cash flow to organic asset expansion.
- Shareholder Yield Champion:** CapEx / CFO $< 30\%$, Dividend Payout $> 50\%$. Mature, cash-rich business returning cash via dividends/buybacks.
- Balanced Compounder:** CapEx / CFO 30-50%, Dividend Payout 20-40%. Funds sustainable growth while paying regular yields.
- De-leveraging Builder:** CFF Net Outflow $> 40\%$ of CFO. Aggressively paying down bank borrowings to de-risk balance sheet.

5. Rule-Based NLP Financial Text Parser

The NLP Engine (src/analysis/nlp_engine.py) processes institutional commentary, concall transcripts, and annual reports to automatically generate structured Pros and Cons matrices for all 92 companies with quantifiable confidence scores.

Information Extraction Pipeline & Heuristics

Extraction Rule	Trigger Keywords & Regex	Category	Confidence
R-01: Debt Reduction	reduced debt, debt free, repaid borrowings	Balance Sheet	92%
R-02: Consistent ROE	ROE > 20%, 3-year track record, high ROCE	Profitability	95%
R-03: Dividend Yield	dividend payout > 30%, healthy dividend	Shareholder Return	88%
R-04: High Valuation	trades at P/B > 10x, elevated P/E multiple	Valuation	85%
R-05: Low Interest Cover	interest coverage < 1.5x, high finance costs	Solvency Risk	90%
R-06: Working Capital Drag	debtor days increased, cash cycle stretched	Operational	84%

Confidence Scoring Methodology

- **Quantitative Verification:** If an extracted qualitative Pro (e.g. 'Company is virtually debt free') matches actual quantitative data ($D/E < 0.1$), confidence score is boosted to 95%.
- **Multi-Sentence Reinforcement:** Repeated thematic mentions across management discussion chapters increase weight and confidence.

6. Machine Learning Clustering & Archetype Profiling

Using unsupervised KMeans machine learning (src/analytics/clustering.py) across 10 normalized financial features, all 92 companies were clustered into 5 distinct behavioral investment archetypes (\$k=5\$, random_state=42).

The 5 Strategic Cluster Archetypes

Cluster Archetype	Count	Dominant Characteristics	Representative Tickers
High-Quality Compounders	51	High ROE/ROCE (>22%), low D/E (<0.3), stable margins	TCS, INFY, TITAN, HINDUNILVR
Turnaround Candidates	16	Volatile PAT, rebounding CFO, restructuring leverage	TATAMOTORS, VEDL, SAIL
Defensive Cash Champions	10	High dividend payout (>50%), recession-proof cash flows	ITC, COALINDIA, POWERGRID
Value Cyclicals	9	Low P/E & P/B multiples, high capital intensity, GDP linkage	TATASTEEL, JSWSTEEL, ONGC
Emerging Growth	6	High revenue CAGR (>18%), high CapEx reinvestment	TRENT, BEL, HAL, DIXON

Mathematical Methodology & Dimensionality Reduction

- **Preprocessing:** Sector-median missing value imputation followed by standard Z-score scaling (StandardScaler).
- **Elbow Method:** Inertia curve evaluated across \$k=2\$ through \$10\$; inflection point at \$k=5\$ minimizes within-cluster sum of squares.
- **PCA Projection:** 2-component Principal Component Analysis capturing > 62% variance for 2D visual cluster boundaries.

7. Production REST API Reference (Endpoints 1–8)

The FastAPI REST server exposes 16 high-performance endpoints mounted under /api/v1. All endpoints return JSON payloads with CORS enabled and include execution timing headers (X-Process-Time-Ms).

#	Method & Endpoint	Parameters	Description & Response Summary
1	GET /api/v1/health	None	System health check with live row counts across all 10 SQLite tables
2	GET /api/v1/companies	limit=20, offset=0, sector	Paginated list of master companies with metadata and counts
3	GET /api/v1/companies/{ticker}	ticker: str	Full company profile, industry, sector, shares outstanding
4	GET /api/v1/companies/{ticker}/pl	ticker: str	10-year historical Profit & Loss statements with OPM and EPS
5	GET /api/v1/companies/{ticker}/bs	ticker: str	10-year historical Balance Sheet records (Assets, Equity, Borrowings)
6	GET /api/v1/companies/{ticker}/cashflow	ticker: str	10-year historical Cash Flow statements (CFO, CFI, CFF, Net Cash)
7	GET /api/v1/companies/{ticker}/ratios	ticker: str	10-year computed financial ratios (ROE, ROCE, D/E, ICR, Net Margin)
8	GET /api/v1/companies/{ticker}/tearsheet	ticker: str	Direct binary streaming download of institutional 2-page PDF

cURL Request Sample: Company Ratios

```
curl -X GET "http://localhost:8000/api/v1/companies/RELIANCE/ratios" -H "Accept: application/json"
```

8. Production REST API Reference (Endpoints 9–16)

Continuation of the API catalog covering multi-slider screener, sector aggregations, peer benchmarking, valuation multiples, portfolio stats, and document retrieval.

#	Method & Endpoint	Parameters	Description & Response Summary
9	GET /api/v1/companies/peers/compare	tickers: str (csv)	Multi-company side-by-side KPI matrix for peer comparison
10	GET /api/v1/screener	min_roe, max_de, sector...	Multi-criteria financial screener with HTTP 400 range validation
11	GET /api/v1/sectors	None	List of all 11 Nifty 100 sectors with company counts and medians
12	GET /api/v1/sectors/{sector}/companies	sector: str	All companies in a sector with 404 validation on unknown sector
13	GET /api/v1/peers/{group_name}	group_name: str	Benchmark peer basket KPIs (e.g. it_majors, private_banks)
14	GET /api/v1/market-cap/{ticker}	ticker: str	Real-time derived market capitalization and valuation metrics
15	GET /api/v1/portfolio/stats	None	Portfolio-wide percentiles (P10, P25, P50, P75, P90) across 10 KPIs
16	GET /api/v1/documents/{ticker}	ticker: str	Annual report links, concall transcripts, and qualitative documents

cURL Request Sample: Screener Query

```
curl -X GET
"http://localhost:8000/api/v1/screener?min_roe=15&max_de=1.0&sector;=Information%20Technology"
```

9. Data Quality Rules & Acceptance Verification

Institutional financial pipelines require uncompromising data integrity. The project enforces 16 automated Data Quality rules (DQ-01 to DQ-16) and 20 Acceptance Gates (AC-01 to AC-20) before deployment.

Core Data Quality Integrity Rules

Rule ID	Rule Name	Severity	Verification Logic
DQ-01	Company PK Uniqueness	CRITICAL	companies.company_id must have zero duplicate values
DQ-02	Annual PK Uniqueness	CRITICAL	(company_id, year) composite key unique across 4 time series tables
DQ-03	Foreign Key Integrity	CRITICAL	Zero orphaned child records in time series tables missing in master
DQ-04	Balance Sheet Balance	WARNING	Total Assets == Total Liabilities + Total Equity (within rounding delta)
DQ-05	OPM Cross-Check	WARNING	Computed Operating Profit Margin matches reported within 1.0%
DQ-06	Positive Sales	WARNING	Reported revenue/sales must strictly exceed zero for active firms
DQ-07	Year Range Format	CRITICAL	All financial years normalized to integer between 2000 and 2030
DQ-08	Ticker Format	CRITICAL	Ticker symbols uppercase, trimmed, alphanumeric characters only

Day 45 Acceptance Gate Sign-Off

- **100% Pass Rate:** All 20 formal acceptance gates (AC-01 to AC-20) verified via automated test harness.
- **Zero Critical DQ Violations:** Zero duplicate primary keys, zero foreign key orphan rows, and zero NaN schema breaches.

10. Financial Ratio Formula & Methodology Reference

Standardized mathematical definitions utilized across all KPI calculators, screener filters, and PDF tearsheets in the Nifty 100 Financial Intelligence Engine.

Financial KPI	Mathematical Formula	Target Benchmark / Norm
Operating Profit Margin (OPM)	$\text{Operating Profit} / \text{Net Sales} * 100$	> 15% (Sector Dependent)
Net Profit Margin (NPM)	$\text{Net Profit after Tax} / \text{Net Sales} * 100$	> 10%
Return on Equity (ROE)	$\text{Net Profit} / (\text{Equity Capital} + \text{Reserves}) * 100$	> 15% (Compounder benchmark)
Return on Capital Employed (ROCE)	$\text{EBIT} / (\text{Total Equity} + \text{Total Borrowings}) * 100$	> 18%
Return on Assets (ROA)	$\text{Net Profit} / \text{Total Assets} * 100$	> 6% (Non-Financial)
Debt-to-Equity (D/E)	$\text{Total Borrowings} / \text{Total Equity}$	< 1.0x (< 0.5x Conservative)
Interest Coverage Ratio (ICR)	$(\text{Operating Profit} + \text{Other Income}) / \text{Interest}$	> 3.0x (Flag if < 1.5x)
Asset Turnover	$\text{Net Sales} / \text{Total Assets}$	> 1.0x
Compound Annual Growth Rate	$((\text{Ending Value} / \text{Starting Value}) ^ {1/n} - 1) * 100$	> 12% across 3Y/5Y cycles
Free Cash Flow (FCF)	$\text{Cash from Operations (CFO)} - \text{Capital Expenditures}$	> 0 (Self-funding business)

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